PHYTOTHERAPY RESEARCH
*Phytother. Res.* **29**: 22–29 (2015)
Published online 30 August 2014 in Wiley Online Library
(wileyonlinelibrary.com) **DOI**: 10.1002/ptr.5216

# Evaluation of Hemostatic and Anti-inflammatory Activities of Extracts from Different *Lagochilus* Species in Experimental Animals: Comparison of Different Extractives and Sources

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**Different members of *Lagochilus* genus have been used in folkloric medicine to treat hemorrhages and inflammation. However, only a few species of them have received scientific attention supporting their efficacy. Here, the hemostatic and antiinflammatory activities of five *Lagochilus* species were determined and compared by using *in vivo* assays. The results showed that the extracts of *Lagochilus lanatonodus* and *Lagochilus diacanthophyllus* showed better hemostatic activities among five species. The high doses of *L. lanatonodus* extracts were able to shorten the values of thrombin time, activated partial thromboplastin time and prothrombin time in a rat model. Moreover, the extracts of *L. lanatonodus* and *L. diacanthophyllus* showed strong inhibitory effects on the acute phase of inflammation in both xylene-induced ear edema mouse model and carrageenan-induced paw edema rat model. In parallel, the treatment of these extracts modulated the expressions of those inflammatory parameters, that is, nitric oxide, prostaglandin $E_2$, inducible nitric oxide synthase, malondialdehyde and superoxide dismutase. *L. lanatonodus* and *L. diacanthophyllus* showed better hemostatic and antiinflammatory activities in several test models: these results therefore supported the folkloric utilization. *L. lanatonodus* was found to be the most active *Lagochilus* species.** 

*Keywords:* hemostatic; antiinflammatory; *Lagochilus* species; traditional herbal medicine.

## INTRODUCTION

The genus *Lagochilus* is a relatively widespread mountain plant of the Lamiaceae family, which contains 35 species and is distributed mainly in central Asia such as Turkistan, Iran, Afghanistan, Russian, Mongolia and China (Rechinger and Hedge, 1982; Yu *et al.*, 1982). Some plants from this genus have been applied for styptic and treatment of allergic dermatosis (Panossian and Wikman, 2008; Zainutdinov *et al.*, 2002) and also been used in preparation on clot retraction (Akopov, 1954) and glaucoma (Kadyroza, 1955). Among different *Lagochilus* species, *Lagochilus inebrians* has long been used as a herbal medicine for its intoxicating and sedative properties by Tajik, Tartar, Turkmen and Uzbek tribesmen (Perry *et al.*, 2002), and its herb named inebrians, or intoxicating mint, refers to its traditional use as an inebriant: its leaves, fruiting tops or flowers are traditionally mixed with honey or sugar to be consumed as tea (Ratsch, 2001). In Russian folk medicine, *L. inebrians* is used to treat allergies, blood clots, skin diseases and hemophilia, and this herb has been listed in the Russian Pharmacopeia as a natural tranquilizer (Ratsch, 2001). Diterpenoids (Zainutdinov *et al.*, 1975; Mavlyankulova *et al.*, 1977; Islamov *et al.*, 1978; Nurmatova *et al.*, 1979), flavonoids (Nasrullaev and Makhsudova, 1991; Gohari *et al.*, 2011), triterpenoids (Gohari *et al.*, 2011) and polysaccharides (Malikova and Rakhimov, 1997) are the constituents isolated from *Lagochilus* species. Pharmacological results indicated that diterpenoids from this genus, mainly lagochiline and its derivatives, could have hemostatic abilities (Zainutdinov *et al.*, 2002). The oil from the leaves and flowers of *Lagochilus kotschyanus* exhibited antibacterial activity (Taban *et al.*, 2009), and anticoagulant activity was reported from the polysaccharides of *Lagochilus usunachmaticus*, which possessed a marked direct-action activity and exceeded heparin during affection (Rakhimov *et al.*, 1997). These species were also traditionally used for their sedative and intoxicating activities.

In China, there are 14 *Lagochilus* species distributed wildly in the northwest region. One of the species, *Lagochilus ilicifolius*, grows widely in Neimeng, Gansu,

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*Received 18 March 2014*
*Revised 10 July 2014*
*Accepted 26 July 2014*

Shanxi and Ningxia of China, whereas the other 13 species are distributed mainly in the Xinjiang region (Yu *et al*., 1982). Indeed, Xinjiang is the main growing area of *Lagochilus* in China. The whole plants of *Lagochilus hirtus* and *Lagochilus platyacanthus* have been employed as styptic folk medicines in Xinjiang, including the treatments of hemorrhage, coronary heart disease, angina pectoris, ulcer, insomnia and amnesia (Xinjiang Institute of Biology, Pedology and Desert Research Institute, 1984). However, there were no reports on pharmacology of *Lagochilus* species distributed in China. Only in 1997 were the chemical constituents of *Lagochilus lanatonodus* (Ba *et al.*, 1997) firstly examined, and recently, two other *Lagochilus* species, namely *Lagochilus leiacanthus* (Furukawa *et al.*, 2011) and *L. ilicifolius* (Li *et al.*, 2012; Qian *et al.,* 2012) were also studied in chemical constituents, and the constituents in these species were reported to contain flavonoids, lignans, iridoids and phenylpropanoids.

In order to improve the utilization of *Lagochilus*, the hemostatic and antiinflammatory properties of extracts from five *Lagochilus* species, including folk medicines *L. hirtus* and *L. platyacanthus* together with another three species *L. lanatonodus*, *L. diacanthophyllus* and *L. ilicifolius*, were firstly reported. The aqueous and ethanol extracts were obtained from these species, which were submitted to biological assays: hemostatic and antiinflammatory activities by using animal models. The current studies were able to verify the traditional medicinal use of these *Lagochilus* species in China.

## MATERIALS AND METHODS

**Plant materials.** The whole plants of *L. ilicifolius* (voucher no. DQY-09072801) were collected from Ningxia China in July 2009, and *L. lanatonodus* (voucher no. MJ-10072301), *L. platyacanthus* (voucher no. KC-10072205), *L. diacanthophyllus* (voucher no. EC-10071401) and *L. hirtus* (voucher no. YM-10072202) were collected from Xinjiang China in July 2010. The voucher specimens were authenticated by Dr Hong Xu, Institute of Chinese Materia Medica, Shanghai University of Traditional Chinese Medicine, Shanghai, China. The voucher specimens were deposited in the Institute of Chinese Materia Medica for further references.

**Reagents.** Yunnan Baiyao was obtained by the Baiyao Group Co., Ltd. (Yunnan, China). Aspirin was produced by Bayer Health Care (Germany). Clopidogrel hydrogen sulfate (CIH) was purchased from the National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). Carrageenan was obtained from Sigma (St. Louis, MO). Xylene and ethanol were obtained from Sinopharm Group Co. Ltd (Shanghai, China); activated partial thromboplastin time (APTT) reagent, prothrombin time (PT) reagent, thrombin time (TT) reagent were obtained from Sysmex Corporation (Fukuoka, Japan). All chemical reagents were of analytical reagent grade.

**Herbal extraction and quality control.** The collected plants were dried under shade at room temperature, and then they were powdered to a homogeneous size by a screen stencil, sieved through a no. 80 mesh. The herbal powders of 200 g were boiled with 2 L of water for 4 h for the first extraction, another 4 h for the second extraction and 3 h for the last extraction, respectively. The extracts were pooled and concentrated to obtain aqueous extracts. For ethanol extracts, the powders of 200 g were submerged with 2 L of 95% ethanol and heated under reflux for three times of extraction at the same time as the aqueous extracts. The combined ethanol extracts were concentrated in a rotary evaporator at reduced pressure, and then a brownish-black-colored residue was obtained. The extracts of five *Lagochilus* herbs were analyzed by HPLC method to control the qualities in our study. The HPLC system was an Agilent Technologies (Santa Clara, CA) 1260 Series consisting of a quaternary pump, continuous vacuum degasser, thermostated auto-sampler and column compartment coupled to a variable wavelength diode-array detector. The samples were separated by a $C_{18}$ column (4.6 mm × 250 mm) with 0.1% aqueous phosphoric acid solution (A)–acetonitrile (B) as mobile phase (for the aqueous extracts, 8% → 12% B at 0–40 min and 12% → 26% B at 40–110 min; for the ethanol extracts, 10%→18% B at 0–40 min, 34% B at 80 min and 54% B at 100 min). The flow rate was 0.8 mL/min with the injection volume (5 μL), and the detection wavelength was set at 310 nm.

**Animals.** The Institute of Cancer Research mice growing for 6–7 weeks at a weight of 20–22 g (for hemostasis and antiinflammatory tests), the Sprague Dawley rats growing for 9–10 weeks at a weight of 250–270 g (for hemostasis test) and the Sprague Dawley rats growing for 4–6 weeks at a weight of 90–110 g (for antiinflammatory test) were purchased from the Chinese Academy of Sciences, Shanghai, and fed with a standard pellet diet, water *ad libitum*. The animals were maintained under standard animal housing conditions (22 ± 3 °C, 40–70% humidity, 12-h light/dark cycle) for 7 days, and then they were randomly allocated to different groups of six animals each and acclimated for 3 days before initiating the pharmacological studies. The experiments were carried out during the light cycle, and the animals were fasted for 12 h before the test. The use of laboratory animals was approved by the Animal Research Ethical Committee of the Shanghai University of Traditional Chinese Medicine. The animal welfare practices and animal experimental protocols were strictly consistent with the Guide for the Care and Use of Laboratory Animals (US National RESEARCH COUNCIL, 2011) and the related ethics regulations of the Shanghai University of Traditional Chinese Medicine.

**Acute toxicity test.** An acute toxicity study was performed in mice according to the Organization for Economic Cooperation and Development (OECD/OCED) guidelines 423 (Acute Oral Toxicity-Acute Toxicity Class Method) (OECD, 2001) in each group of ten mice. The animals were orally administrated a dose of 5000 mg/kg aqueous and ethanol extracts from five *Lagochilus* species and were observed continuously for 30 min and for 4 h, intermittently for 24 h and then once a day for the next 14 days for general behavioral change, signs of toxicity and mortality.

**Design of drug treatment.** The animals were divided into the following groups (six animals for each group): normal control group [received physiological saline for 7 days per os (p.o.)], positive control group (received Yunnan Baiyao and CIH for 7 days p.o. for hemostasis test and aspirin at the seventh day p.o. for anti-inflammatory test) and extract group (received aqueous and ethanol extracts at doses of 200 and 400 mg/kg separately for 7 days p.o.). Yunnan Baiyao, CIH and aspirin were respectively dissolved in the same vehicle and given orally at doses of 200, 13.5 and 200 mg/kg. The aqueous and ethanol extracts from five *Lagochilus* species were administered to the animals at doses of 200 or 400 mg/kg after being suspended in physiological saline.

**Hemostasis assay.** The procedure used to assess the pro-coagulant properties of different extracts was according to previous reports with modifications (Xu *et al.*, 1991; Chen, 1993). Ethanol and aqueous extracts, Yunnan Baiyao, CIH and the equivalent physiological saline were given orally for 7 days continuously. After 45 min of the last administration, the blood was collected using a capillary vessel from the inner canthus of the eye, and the time required for clotting strands to emerge in both ends was recorded as blood clotting time (Li *et al.*, 2008). In the tail bleeding test, the method of Kim and Lee (2006) with mild corrections was followed. The mouse tail was transected at 3 mm from the tip after 45 min of the last administration, and the maimed tail was placed into a tube containing 8 mL physiological saline at 37 °C. The interval, from the tail transection until the time that the blood was no longer apparently overflowing, was measured as bleeding time.

**Measurement of hemostatic parameters.** To investigate the primary mechanism of antihemorrhage, the hemostatic parameters were measured. Indexes were justified after 45 min of the last administration, then rats were anesthetized with urethane and sampled from the abdominal aorta with anticoagulation tubes (with the addition of 3.8% sodium citrate 1:9, v/v), the mixture was centrifuged at 3000 rpm for 10 min and the rat plasma was obtained for the assessment of blood clotting indexes using a Sysmex CA-530 coagulation analyzer (Sysmex Corporation, Fukuoka, Japan).

**Antiinflammatory assay.** In the xylene-induced ear edema in mice, the antiinflammatory property of the extracts was confirmed using the method described by Kou *et al.* (2005) with some modifications. The mean edema inhibition percentage (%) was calculated using the following formula: inhibition rate $(\%) = (B - A)/B \times 100$, where $A$ is the difference of ear edema weight in the extract/aspirin-treated group and B is the difference of ear edema weight in the normal control group.

In the carrageenan-induced paw edema in rats, the methods of Yesilada and Küpeli (2002) with minor alterations were followed. The plantar injection of freshly prepared carrageenan suspension (1%, 0.1 mL) was carried out 1 h after the last administration of extracts, vehicle and aspirin. The paw volume was measured immediately before and after 1% carrageenan treatment at 1, 3 and 5 h by an electric plethysmometer, but the mechanism was explained only for the third hour at which the maximum edema occurred. The results were presented as the volume edema (mL) in relation to the value obtained at time 0 h. The inhibition percentage (%) was calculated according to the formula given in the following: (control group mean − test group mean)/control group mean × 100.

**The assays of prostaglandin $E_2$, nitric oxide, inducible nitric oxide synthase, superoxide dismutase and malondialdehyde.** The serum was collected from the eye socket of rats at the third hour for the determination of the levels of prostaglandin $E_2$ ($PGE_2$), nitric oxide (NO), inducible NO synthase (iNOS), malondialdehyde (MDA) and superoxide dismutase (SOD): the assays were performed using the kits for $PGE_2$, NO, iNOS, MDA and SOD (Nanjing Jiancheng Bioengineering Institute, China) based on the manufacturer's instructions.

**Data analysis.** A software program, SPSS version 15.0 (SPSS Inc., Chicago, IL), and descriptive statistics were used to describe the data, which were presented as mean ± SD. Data obtained were initially examined to confirm normal distribution and homogeneity of

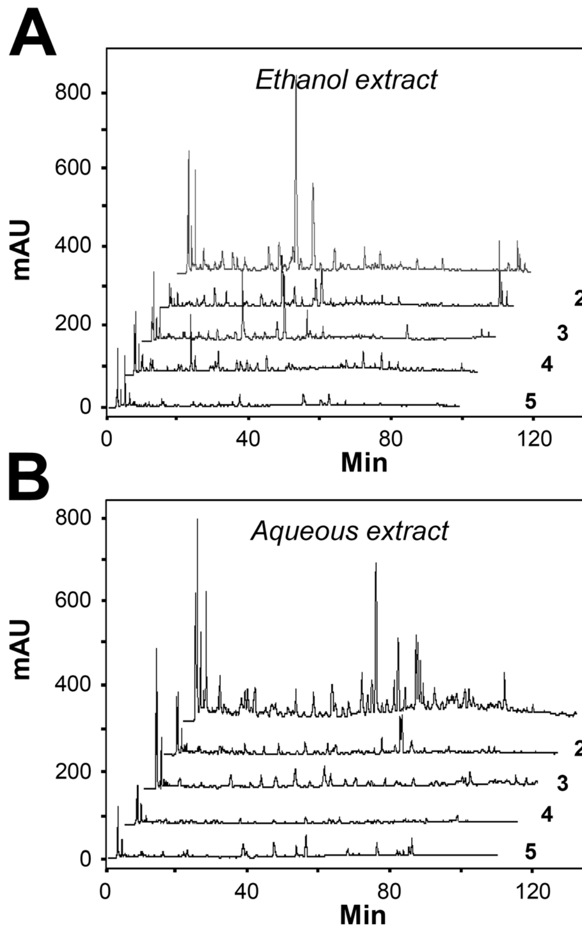


**Figure 1.** HPLC chromatograms of ethanol (A) and aqueous extracts (B) from five *Lagochilus* species. More peaks were found in aqueous extracts, and significant differences were shown among extracts from different species in their fingerprints, indicating the HPLC fingerprints could serve as an efficient quality control tool for *Lagochilus* herbal extracts. (1: *Lagochilus platyacanthus*; 2: *Lagochilus ilicifolius*; 3: *Lagochilus hirtus*; 4: *Lagochilus diacanthophyllus*; 5: *Lagochilus lanatonodus*).

variance. Analysis of variance was used to compare different variables between the groups followed by Dunnett's test. Values of $P < 0.05$ were taken to imply statistical significance ($^*P < 0.05$; $^{**}P < 0.01$).

## RESULTS

### Acute toxicity studies

Oral administration of *Lagochilus* extracts, including aqueous and ethanol extracts from five species, did not show visible signs of toxicity in mice at the highest dose of 5000 mg/kg of body weight. There was no evidence for differences in physiological or behavioral responses between the normal control group and extract-treated groups and also no differences in the consumption of food and water, which implied that the five *Lagochilus* species are not toxic.

### Standardization of *Lagochilus* extracts

Both aqueous and ethanol extracts were obtained from the five chosen *Lagochilus* herbs. The extracts from five individual batches of each *Lagochilus* species were performed: this was to ensure the repeatability of extracts. The yield of each extraction was calibrated with aqueous extracts of 14.62%, 21.23%, 25.05%, 23.99% and 24.56% and ethanol extracts of 5.27%, 9.03%, 8.81%, 8.27% and 7.38% for *L. lanatonodus*, *L. diacanthophyllus*, *L. platyacanthus* *L. hirtus* and *L. ilicifolius*, respectively. The total yield of aqueous extraction was much better than that of ethanol extraction. In addition, *L. lanatonodus* had a lower yield of extraction in both extracting solvents. The extracts were subjected to HPLC fingerprint analyses. As expected, the aqueous extracts showed more peaks than did ethanol extracts (Fig. 1A and B). In addition, no two extracts showed similarity in the fingerprints. Both extraction yield and fingerprint could serve as quality control for the herbal extracts.

### Effects on blood clotting time and tail bleeding time

In order to assess the hemostatic activity of the extracts from *Lagochilus* species, mice were orally administered with different extracts at a dose of either 200 or 400 mg/kg. The effects on blood circulation, including blood clotting and tail bleeding time, were subsequently determined. CIH (13.5 mg/kg), which was used as a control drug, prolonged the bleeding time and clotting time. A well-known traditional Chinese hemostatic medicine, Yunnan Baiyao, was included for comparison with *Lagochilus* species. As shown in Fig. 2A, the blood clotting time of mice, treated with the extracts, showed a reduction of 20–40% under high doses of *Lagochilus*

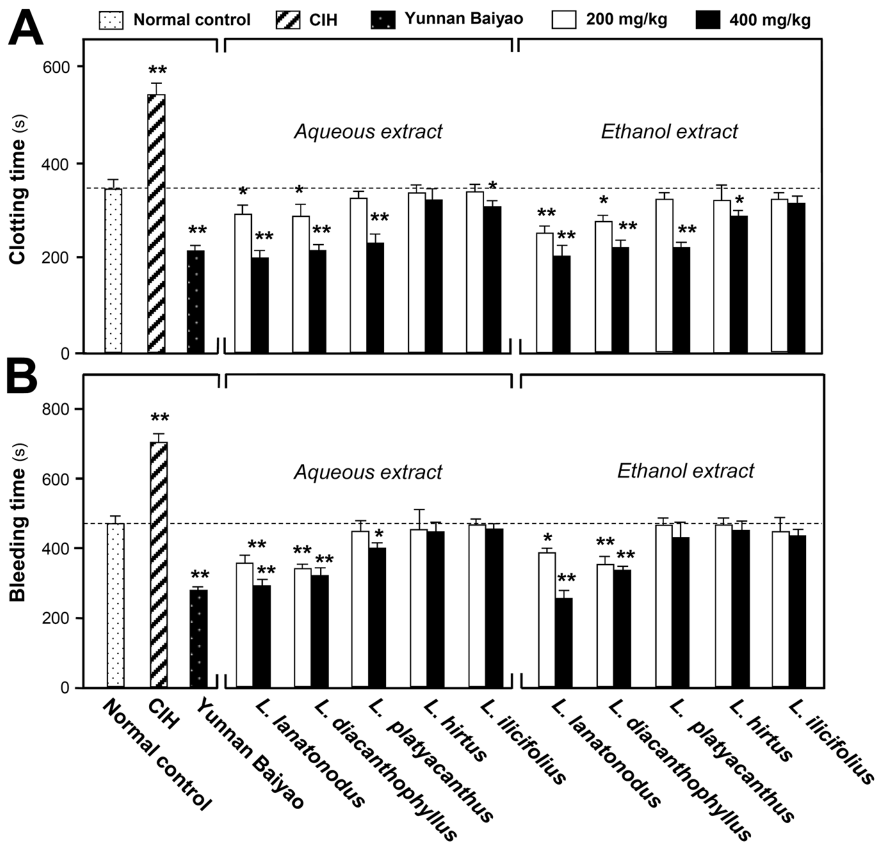


**Figure 2.** Effect of different extracts from five *Lagochilus* species on clotting time (A) and bleeding time (B) in mice. Ethanol and aqueous extracts (200 or 400 mg/kg), Yunnan Baiyao (200 mg/kg), clopidogrel hydrogen sulfate (CIH; 13.5 mg/kg) and the equivalent physiological saline were given orally for 7 days continuously. After 45 min of the last administration, the clotting time was determined by using the capillary method. Extracts from *Lagochilus lanatonodus* and *Lagochilus diacanthophyllus* significantly reduced the bleeding time in a dose-dependent manner, which was comparable with the effect of Yunnan Baiyao. The bleeding time was determined by cutting tail. Extracts from *L. lanatonodus* and *L. diacanthophyllus* significantly inhibit the bleeding time. Ethanol extract of *L. lanatonodus* at the 400-mg/kg dose showed the strongest inhibition of about 48%, which is even stronger than that of Yunnan Baiyao. Values are mean ± SD ($n$ = 6), $^*P < 0.05$, $^{**}P < 0.01$ compared with the normal control group (Dunnett's test).

 

extracts. The extracts of *L. ilicifolius* and *L. hirtus* showed very minor effects. In this assay, the best extracts were identified from *L. lanatonodus* and *L. diacanthophyllus*, which reduced the bleeding time in a dose-dependent manner (Fig. 2A). The aqueous and ethanol extracts of *L. lanatonodus* at 400 mg/kg showed stronger activities with about 40% inhibition on the clotting time: this effect was better than that of Yunnan Baiyao. Similar effects were observed in bleeding time of the mice tail after the treatments of *Lagochilus* extracts (Fig. 2B). Again, the extracts of *L. lanatonodus* and *L. diacanthophyllus* showed the strongest activity with about 48% inhibition of bleeding time (Fig. 2B). The other extracts also inhibit the bleeding time at a lower degree, except the extracts of *L. ilicifolius* and *L. hirtus*, which showed no reduction (Fig. 2B). In the preceding studies, the extracts of *L. lanatonodus* showed a stronger activity than other extracts in inhibiting the blood clotting and tail bleeding time, and therefore, this herb was selected for further analysis.

## Effects on hemostatic parameters

To assess the effect on rat plasma coagulation and study the potential hemostatic mechanism, *L. lanatonodus* extracts, including aqueous and ethanol extracts, were administered to rats for 7 days continuously. The arterial blood was collected from the iliac bifurcation of the extract-treated rats, and the hemostatic parameters, that is, TT, APTT and PT, were determined by conventional clinical procedures. CIH prevented platelets from aggregating and was used as a control for the hemostatic parameters. As shown in Fig. 3, the aqueous and ethanol extracts of *L. lanatonodus* at 400 mg/kg significantly reduced the value of TT, which was comparable with that of Yunnan Baiyao, although both extracts at 400 mg/kg were also found to decrease the values of APTT and PT at lower extents.

## Effects on xylene-induced ear edema in mice

In order to study the antiinflammatory activities of *Lagochilus* extracts, the aqueous and ethanol extracts of different herbs were compared according to the effect on xylene-induced ear edema in mice. As shown in Fig. 4A, the extracts of *L. lanatonodus*, *L. diacanthophyllus* and *L. platyacanthus* caused a dose-dependent inhibition of the development of ear edema. The extracts from these three species at 400 mg/kg showed significant inhibition on the development of ear edema; for example, the aqueous and ethanol extracts of *L. lanatonodus* at 400 mg/kg induced about 43.1% and 35.1% inhibition, respectively. However, the extracts of *L. hirtus* and *L. ilicifolius* showed no activity.

## Effects on carrageenan-induced paw edema in rats

The extracts of *L. lanatonodus* were selected for further study. In this part, their activities against carrageenan-induced paw edema were evaluated. As shown in Fig. 4B, most of the groups could reduce carrageenan-induced paw edema. The reduction was revealed throughout the entire period of observation. These effects of *Lagochilus* extracts were less than those of aspirin (Fig. 4B). The

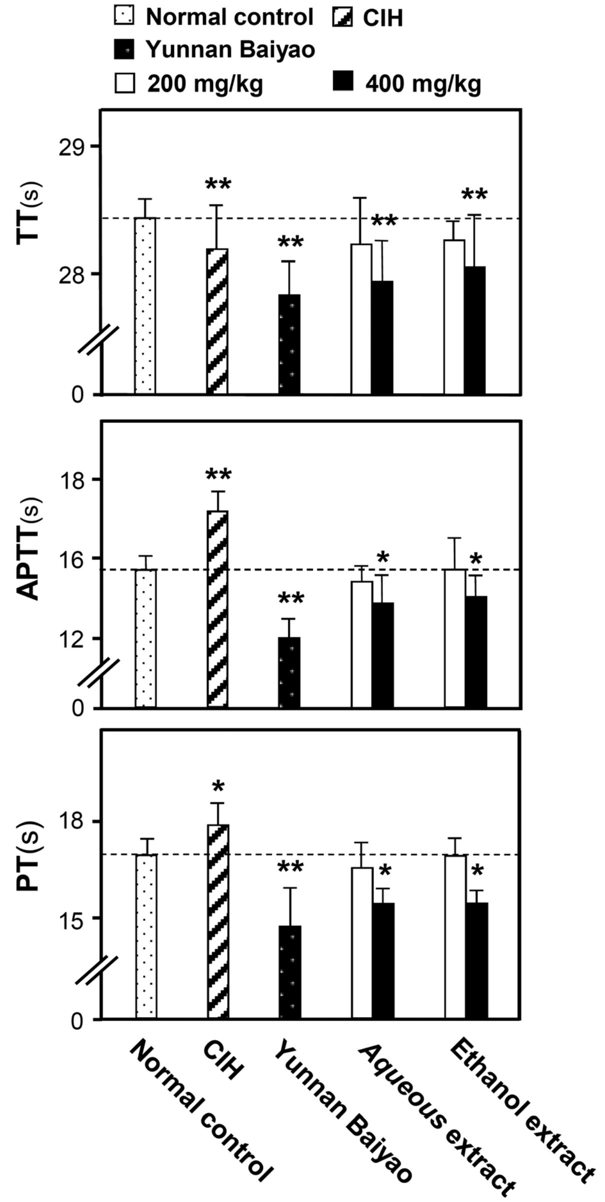


**Figure 3.** Hemostatic parameters, thrombin time (TT), activated partial thromboplastin time (APTT) and prothrombin time (PT) in rats dosed with different extracts from *Lagochilus lanatonodus*. Ethanol and aqueous extracts from *L. lanatonodus* (200 or 400 mg/kg), Yunnan Baiyao (200 mg/kg), clopidogrel hydrogen sulfate (CIH; 13.5 mg/kg) and the equivalent physiological saline were given orally for 7 days continuously. After 45 min of the last administration, arterial blood was collected, and hemostatic parameters TT, APTT and PT were determined by conventional clinical procedures. Both ethanol and aqueous extracts at a 400-mg/kg dose, as well as Yunnan Baiyao (positive control), obviously decreased TT, APTT and PT. Values are mean ± SD ($n = 6$), $*P < 0.05$, $**P < 0.01$ compared with the normal control group (Dunnett's test).

aqueous and ethanol extracts at a dose of 400 mg/kg reduced the edema volume after 3 h of treatment, having inhibition ratios of 41.4% and 22.5%, respectively. Meanwhile, the aqueous extract at 200 mg/kg produced a relatively significant inhibitory effect with an inhibition of 7.75%, 3 h after the treatment.

## Effects on the production of inflammatory indexes

To assess the effect of *L. lanatonodus* extracts in carrageenan-induced paw edema, the levels of inflammatory indexes,

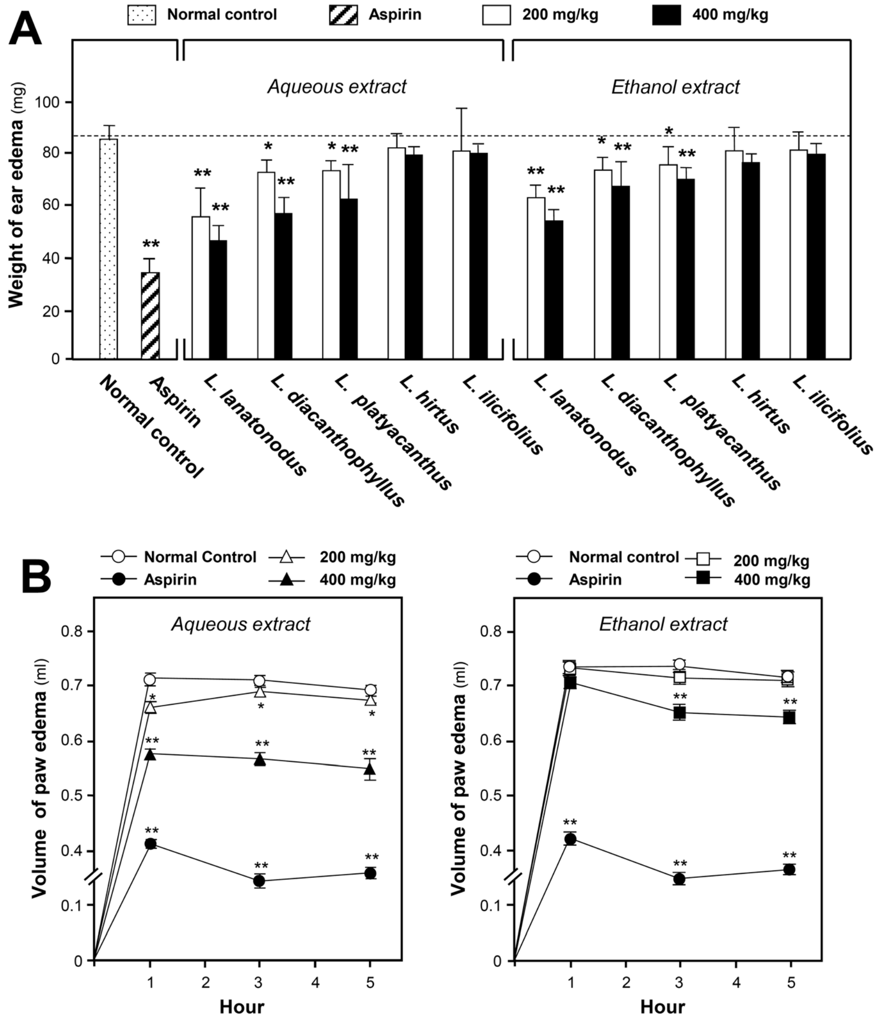


**Figure 4.** (A) Effect of different extracts from five *Lagochilus* species on xylene-induced ear edema in mice. Ethanol and aqueous extracts were given orally for 7 days continuously, whereas aspirin (200 mg/kg) was given once at the last day. Xylene was applied after 45 min of the last administration, and ear edema was evaluated after 15 min of applying xylene. The extracts of *Lagochilus lanatonodus*, *Lagochilus diacanthophyllus* and *Lagochilus platyacanthus* at both 200- and 400-mg/kg doses significantly inhibited ear edema induced by xylene in a dose-dependent manner. And *L. lanatonodus* showed the strongest antiinflammatory effect. Values are mean ± SD ($n = 6$), $^*P < 0.05$, $^{**}P < 0.01$ compared with the control group (Dunnett's test). (B) Effect of different extracts from *L. lanatonodus* on carrageenan-induced paw edema in rats. Ethanol and aqueous extracts from *L. lanatonodus* (200 and 400 mg/kg) were given orally for 7 days continuously, whereas aspirin (200 mg/kg) was given once at the last day. Carrageenan was injected after 1 h of the last administration, and paw edema was evaluated after 1, 3 and 5 h of injecting carrageenan. Reduction of paw edema was revealed throughout the entire period of observation. And the strongest reduction was observed after 3 h of injecting carrageenan for both ethanol and aqueous extracts at 400 mg/kg, with inhibition rates at 41% and 23%, respectively. Values are mean ± SD ($n = 6$). $^*P < 0.05$, $^{**}P < 0.01$ compared with control group (Dunnett's test).

including $PGE_2$, NO, iNOS, MDA and SOD, were determined in the rat serum. As shown in Table 1, the treatments with both aqueous and ethanol extracts of *L. lanatonodus* were able to modulate the inflammatory parameters. High doses of aqueous extracts showed modulation on the expressions of $PGE_2$, NO, iNOS, MDA and SOD: a lower dose was also revealed in modulating NO, iNOS and MDA. In parallel, the ethanol extracts at high doses showed modulation on all the tested parameters, except the level of MDA. Aspirin, serving as a control, showed positive effects in all cases (Table 1).

## DISCUSSION

*Lagochilus* genus has been widely used as a medical herb in countries in central Asia such as Tajik, Tartar and Russia. Some species were officially recorded in the pharmacopeia for their versatile pharmacological effects. However, only a few species have been scientifically investigated for their efficacy. On the other hand, 14 *Lagochilus* species were distributed in the northwest region of China, among which 13 species are distributed in Xinjiang and employed as a folk medicine. But there were no pharmacological reports on them. Therefore, in this study, the hemostatic and antiinflammatory activities of five *Lagochilus* species in China were investigated, for better utilization of *Lagochilus*.

The extracts of *L. lanatonodus* and *L. diacanthophyllus* showed dose-dependent hemostatic effects, whereas the other three species showed little or no effects. Both aqueous and ethanol extracts of *L. lanatonodus* at 400 mg/kg greatly reduced the blood clotting time and tail bleeding time, which were comparable with, or even better than, the effect of Yunan Baiyao, a golden standard of herbal formula for hemostatic. In addition, high doses of *L. lanatonodus* extracts, both aqueous and

**Table 1. Effects of *Lagochilus lanatonodus* extracts on the levels of $PGE_2$, NO, iNOS, MDA and SOD in carrageenan-induced paw edema rat model**

| Group | Dose (p.o.) | Number of animal | $PGE_2$ (ng/L) | NO (μmol/L) | iNOS (U/mL) | MDA (μmol/L) | SOD (U/mL) |
|---|---|---|---|---|---|---|---|
| Control | 1 mL/100 g | 6 | 0.241 ± 0.023 | 32.39 ± 0.75 | 94.53 ± 0.71 | 32.90 ± 0.77 | 3.65 ± 0.097 |
| Aspirin | 200 mg/kg | 6 | 0.175 ± 0.015** | 26.66 ± 0.83** | 89.16 ± 0.74** | 27.25 ± 0.70** | 4.15 ± 0.096** |
| Aqueous | 200 mg/kg | 6 | 0.220 ± 0.015 | 30.15 ± 0.76** | 92.12 ± 0.65** | 31.74 ± 0.82* | 3.77 ± 0.093 |
| extract | 400 mg/kg | 6 | 0.197 ± 0.023* | 28.70 ± 0.63** | 90.41 ± 0.78** | 29.25 ± 0.65** | 3.96 ± 0.097** |
| Ethanol | 200 mg/kg | 6 | 0.212 ± 0.018 | 30.71 ± 0.79* | 93.43 ± 0.87 | 32.07 ± 0.63 | 3.77 ± 0.064 |
| extract | 400 mg/kg | 6 | 0.205 ± 0.017* | 29.53 ± 0.89** | 91.14 ± 0.84** | 31.90 ± 0.67 | 3.97 ± 0.071** |

Dunnett's test: serum was obtained at the third hour at which the maximum edema occurred and determined for inflammatory factors. Values are mean ± SD ($n$ = 6). iNOS, inducible nitric oxide synthase; MDA, malondialdehyde; NO, nitric oxide; $PGE_2$, prostaglandin $E_2$; p. o., per os; SOD, superoxide dismutase.

*$P < 0.05$,

**$P < 0.01$, compared with control group.

ethanol extracts, reduced the values of blood coagulation factors, for example, TT, APTT and PT. Thus, the hemostasis of *L. lanatonodus* could be mainly implemented by affecting the conversion of fibrinogen to fibrin.

The antiinflammatory property is another major activity of *Lagochilus* species. Xylene-induced ear edema and carrageenan-induced paw edema are two well-characterized models for the study of inflammation and are frequently used to evaluate the anti-inflammatory effects of natural products and have good predictive value for screening antiinflammatory agents (Mujumdar and Misar, 2004; Busnardo *et al*., 2010). The former reflects the edematization, whereas the latter mimics the prototype of the exudative phase. Here, the aqueous and ethanol extracts of *L. lanatonodus* showed stronger activation than did the other four species. Meanwhile, the extracts of *L. diacanthophyllus* and *L. platyacanthus* also significantly inhibited the development of ear edema induced by xylene at a weaker degree. Moreover, the aqueous and ethanol extracts of *Lagochilus* showed a reduction of paw edema throughout the entire period. The strongest inhibition was observed at the third hour at a dose of 400 mg/kg.

The inflammatory indexes, including $PGE_2$, NO, iNOS, MDA and SOD, were regulated by the treatment of *L. lanatonodus* extracts. Among the five inflammatory factors, iNOS and cyclooxygenase (COX) are two key enzymes regulating the production of NO and $PGE_2$, which are the central indexes of inflammation (Posadas *et al*., 2000). iNOS is rapidly induced by stimuli such as cytokines and produces large amounts of NO (Wang and Marsden, 1995): NO is an essential bioregulatory molecule required for several physiological processes like neural signal transmission, immune response, vasodilation and control of blood pressure. An increased level of NO derived from iNOS can result in the formation of peroxynitrite after a reaction with oxygen free radicals during inflammatory responses (Posadas *et al*., 2000). $PGE_2$ is the major metabolite produced by COX-2 at inflammation sites, which contributes to edema and pain sensitization and increases local blood flow (Giuliano and Warner, 2002). MDA is a sign to express the level of lipid peroxidation, that is, reflect the injury degree made by irritants indirectly (Gaweł *et al*., 2004). Thus, a drug capable of preventing the release of these four inflammatory mediators should potentially possess antiinflammatory activities. SOD is able to eliminate hazardous substance produced in the metabolism of organisms (Yasui and Baba, 2006). In the present study, the *Lagochilus* extracts, especially at the 400-mg/kg dose, reduced NO production, lowered the level of MDA and increased the level of SOD, suggesting that they could inhibit the lipid peroxidation and scavenge radicals by enhancing the activities of the antioxidant enzymes. Besides, decreased $PGE_2$ level, triggered by the treatment of *L. lanatonodus* extracts could be related with the suppression of synthesis and/or release of endogenous pro-inflammatory substances.

## CONCLUSION

Taken together, *L. lanatonodus* showed the best hemostatic and antiinflammatory activities among the five *Lagochilus* species. The aqueous extract of *L. lanatonodus* showed better pharmacological effects than the ethanol extract. Besides, *L. diacanthophyllus* also showed weaker hemostatic and antiinflammatory activities than did *L. lanatonodus*. However, *L. platyacanthus*, *L. hirtus* and *L. ilicifolius* exhibited little potency in relieving bleeding and alleviating edema. *L. lanatonodus* extracts can significantly reduce TT and modulate inflammatory indexes, that is, lower NO, MDA and $PEG_2$ and elevate SOD. Although other factors might also be responsible for these activities, it is a first scientific support for the folkloric usage of *Lagochilus* species for treatment of hemorrhage and antiinflammation diseases in China. The comparison of the five species also provided a basis for the better utilization of herbal medicines from the *Lagochilus* genus.

### Acknowledgements

This research was supported by a grant (no. 09405801700) from the Science and Technology Commission of Shanghai Municipality.

### Conflict of Interest

The authors have declared that there is no conflict of interests.

## REFERENCES

Akopov IE. 1954. Effect of preparation of *Lagochilus* on clot retraction. *Biull Eksp Biol Med* **37**: 49–54.

Ba H, Tolhen, Musadillin S, Wang BD, *et al*. 1997. Studies on the chemical constituents of *Lagochilus lanatonodus*. *Nat Prod Res Dev* **9**: 44–48.

Busnardo TC, Padoani C, Mora TC, *et al*. 2010. Anti-inflammatory evaluation of *Coronopus didymus* in the pleurisy and paw edema models in mice. *J Ethnopharmacol* **128**: 519–525.

Chen Q. 1993. Pharmacological Experiment Methodology of Traditional Chinese Medicine (2nd edn). People's Medical Publishing House: Beijing.

Furukawa M, Suzuki H, Makino M, *et al*. 2011. Studies on the constituents of *Lagochilus leiacanthus* (Labiatae). *Chem Pharm Bul* **59**: 1535–1540.

Gaweł S, Wardas M, Niedworok E, *et al*. 2004. Malondialdehyde (MDA) as a lipid peroxidation marker. *Wiad Lek* **57**: 453–455.

Giuliano F, Warner TD. 2002. Origins of prostaglandin $E_2$: involvements of cyclooxygenase (COX)-1 and COX-2 in human and rat systems. *J Pharmacol Exp Ther* **303**: 1001–1006.

Gohari A, Barari E, Saeidnia S, *et al*. 2011. Phytochemical study of *Lagochilus cabulicus* Benth. *Planta Med* **1**: 77–83.

Islamov R, Zainutdinov UN, Aslanov KA. 1978. Lagochilin 3-monoacetate from *Lagochilus inebrians*. *Chem Nat Comp* **14**: 342–343.

Kadyroza KK. 1955. Lagochilus in the treatment of glaucoma. *Vestn Oftalmol* **34**: 32–35.

Kim MS, Lee KA. 2006. Antithrombotic activity of methanolic extract of *Umbilicaria esculenta*. *J Ethnopharmacol* **105**: 342–345.

Kou JP, Sum Y, Lin YW, et al. 2005. Anti-inflammatory activities of aqueous extract from *Radix Ophiopogon japonicus* and its two constituents. *Bio Pharm Bull* **28**: 1234–1238.

Li GZ, Dumaa M, Pu X, *et al*. 2012. Chemical components of aerial part of *Lagochilus ilicifolius*. *Chin J Appl Enviro Bio* **18**: 924–927.

Li MX, Jia ZP, Hu ZD, *et al*. 2008. Experimental study on the hemostatic activity of the Tibetan medicinal herb *Lamiophlomis rotata*. *Phytother Res* **22**: 759–765.

Malikova MK, Rakhimov DA. 1997. Plant polysaccharides VIII. Polysaccharides of *Lagochilus zeravschanicus*. *Chem Nat Comp* **33**: 438–440.

Mavlyankulova ZI, Zainutdinov UN, Aslanov KA. 1977. Diterpenes of *Lagochilus pubescens*. *Chem Nat Comp* **13**: 39–41.

Mujumdar AM, Misar AM. 2004. Anti-inflammatory activity of *Jatropha curcas* roots in mice and rats. *J Ethnopharmacol* **90**: 11–15.

Nasrullaev FD, Makhsudova BT. 1991. Flavonoids of *Lagochilus platycalyx*. *Chem Nat Comp* **27**: 511–512.

Nurmatova MP, Zainutdinov UN, Kamaev FG, *et al*. 1979. Structure and configuration of a new diterpenoid lactone from *Lagochilus hirsutissimus*. *Chem Nat Comp* **15**: 695–699.

OECD. 2001. OECD Guidelines for Acute Toxicity of Chemicals (No. 420). Organisation for Economic Co-operation and Development: Paris.

Panossian A, Wikman G. 2008. Pharmacology of *Schisandra chinensis* Bail: an overview of Russian research and uses in medicine. *J Ethnopharmacol* **118**: 183–212.

Perry E, Ashton H, Young A. 2002. Neurochemistry of Consciousness: Neurotransmitters in Mind. Johan Benjamins Publishing: Amsterdam.

Posadas I, Terencio MC, Guillén I, *et al*. 2000. Co-regulation between cyclo-oxygenase-2 and inducible nitric oxide synthase expression in the time-course of murine inflammation. *Naunyn Schmiedebergs Arch Pharmacol* **361**: 98–106.

Qian JS, Zhang BF, Wang W, *et al*. 2012. Chemical constituents of *Lagochilus ilicifolius*. *Chin Trad Herbal Drugs* **43**: 869–872.

Rakhimov DA, Malikova MK, Vakhabov AA, *et al*. 1997. Plant polysaccharides IX. Isolation and anticoagulant activity of the polysaccharides of *Lagochilus usunachmaticus*. *Chem Nat Comp* **33**: 534–535.

Ratsch C. 2001. Marijuana Medicine—A World Tour of the Healing and Visionary Powder of *Cannabis*. Healing Arts Press: Vermont Rochester.

Rechinger KH, Hedge IC. 1982. Flora Iranica, *Labiatae*. Akademische Druck-u.Verlagsanatalt: Graz.

Taban S, Masoudi S, Chalabian F, *et al*. 2009. Chemical composition and antimicrobial activities of the essential oils from flower and leaves of *Lagochilus kotschyanus* Boiss. A new species from Iran. *J Med Plants* **8**: 58–63.

US National Research Council. 2011. Guide for the Care and Use of Laboratory Animals (8th edn). National Academy Press: Washington DC.

Wang Y, Marsden PA. 1995. Nitric oxide synthases: gene structure and regulation. *Adv Pharmacol* **34**: 71–90.

Xinjiang Institute of Biology, Pedology and Desert Research Institute. 1984. Xinjiang Medicinal Flora (Vol. **3**). Xinjiang People's Publishing House: Xinjiang.

Xu SY, Bian RL, Chen X. 1991. Pharmacological Experiment Methodology (2nd edn). People's Medical Publishing House: Beijing.

Yasui K, Baba A. 2006. Therapeutic potential of superoxide dismutase (SOD) for resolution of inflammation. *Inflamm Res* **55**: 359–363.

Yesilada E, Küpeli E. 2002. *Berberis crataegina* DC. root exhibits potent anti-inflammatory, analgesic and febrifuge effects in mice and rats. *J Ethnopharmacol* **79**: 237–248.

Yu DJ, Wu ZY, Cui HB. 1982. Flora of China. Vol II (2). Science Press: Beijing; 525.

Zainutdinov UN, Mavlyankulova ZI, Aslanov KA. 1975. A chemical study of *Lagochilus pubescens*. *Chem Nat Comp* **11**: 287–288.

Zainutdinov UN, Islamov R, Dalimov DN, et al. 2002. Structure–activity relationship for hemostatic lagochilin diterpenoids. *Chem Nat Comp* **38**: 161–163.